📊 Financial Report Analyzer

A one-day hackathon-style Python project for intelligent financial document analysis

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1. Project Overview

1.1 Value Proposition

The **Financial Report Analyzer** enables rapid analysis of long financial reports by combining PDF parsing, NLP-based compression, sentiment analysis, trend detection, and visualization into a single interactive dashboard.
It reduces analysis time from hours to minutes while preserving key insights.

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2. Problem Statement

Annual and quarterly financial reports typically span **100+ pages**, containing dense narratives, tables, and disclosures. These reports are:

- Time-consuming to read manually
- Difficult to compare across companies and time periods
- Challenging to analyze consistently for risks, sentiment, and trends

As a result, investors, analysts, students, and researchers face delays in extracting actionable insights.
This project addresses the problem by **automating extraction, compression, analysis, and visualization** of financial reports.

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3. Solution Overview

The **Financial Report Analyzer** is a lightweight Python application designed for fast, exploratory financial analysis.

The system performs the following:

- Accepts **uploaded financial report PDFs** (annual or quarterly)
- Extracts **unstructured text and structured financial tables**
- Compresses long narratives using a **ScaleDown-style summarization pipeline**

- Performs **sentiment analysis** on management discussion and transcripts
- Extracts **risk-related disclosures**
- Fetches **stock price trends and peer data** from Yahoo Finance
- Analyzes **earnings call transcripts** (optional input)
- Visualizes insights using a **Streamlit-based dashboard**
- Estimates **ROI and latency savings** achieved through document compression

The scope is intentionally limited to **1-day hackathon feasibility**, prioritizing clarity, modularity, and core functionality.

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4. Key Features

4.1 Document Processing

- PDF financial report parsing
- Narrative compression targeting ~80% reduction
- Financial table extraction from PDFs

4.2 NLP & Intelligence

- Sentiment analysis of financial narratives
- Risk factor detection
- Earnings call transcript analysis

4.3 Financial Analysis

- Stock trend detection using Yahoo Finance
- Peer company comparison
- Rule-based alert generation

4.4 Analytics & Visualization

- ROI and latency savings calculator
- Interactive Streamlit dashboard

5. Technology Stack

Category	Tools
Language	Python
UI	Streamlit
PDF Text Extraction	pdfplumber
PDF Table Extraction	camelot
Data Processing	pandas
Market Data	yfinance
NLP Models	transformers
ML Utilities	scikit-learn
Visualization	matplotlib, plotly

System Architecture

6.1 High-Level Pipeline

User Uploaded PDF
↓
Text & Table Extraction
↓
Narrative Compression
↓
NLP Analysis (Sentiment & Risk)
↓
Market Trend Data (Yahoo Finance)
↓
Interactive Streamlit Dashboard

7. Application Workflow

Step-by-Step Execution

1. User uploads a financial report PDF
2. System extracts text and financial tables
3. Long narratives are compressed
4. Sentiment and risk analysis is performed

5. Stock price and trend data is fetched from Yahoo Finance
6. Peer companies are compared
7. Alerts are generated based on predefined rules
8. Results are displayed in the Streamlit dashboard

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8. Dashboard Components

8.1 Filing Summary

- Compressed overview of the uploaded report
- Generated using ScaleDown-style summarization
- Highlights key financial and business insights

8.2 Risk Factors

- Automatically extracted risk-related statements
- Flags operational, financial, and market risks
- Helps identify potential red flags quickly

8.3 Financial Tables

- Parsed financial tables (income statement, balance sheet, etc.)
- Cleaned and displayed in structured format
- Enables comparison across reporting periods

8.4 Trends

- Stock price and volume trends from Yahoo Finance
- Time-series visualizations
- Identifies upward or downward momentum

8.5 Peer Comparison

- Compares selected company against peer tickers

- Uses market trends and financial indicators
- Supports relative performance evaluation

8.6 Earnings Transcript Analysis

- Sentiment analysis of earnings call transcripts
- Detects management tone (positive, neutral, negative)
- Highlights sentiment shifts across quarters

8.7 ROI Calculator

- Estimates benefits of document compression
- Displays token reduction and latency savings
- Demonstrates efficiency gains from NLP compression

8.8 Alerts

- Rule-based alert generation
- Example triggers include:
    - Negative sentiment
    - High risk density
    - Declining stock trends

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9. Limitations

The current implementation has the following limitations:

1. **Simplified Summarization**
    - Heuristic or general-purpose NLP models
    - May miss fine-grained financial or legal nuances
2. **PDF Parsing Variability**
    - Extraction accuracy depends on PDF formatting
    - Scanned or poorly structured PDFs may reduce quality

3. **Limited Financial Metrics**
    - Focuses on trends and comparisons
    - Does not compute advanced ratios or forecasts
4. **Manual Transcript Input**
    - Earnings transcripts must be pasted manually
    - No automated transcript ingestion
5. **Basic Alert Logic**
    - Rule-based alerts only
    - No adaptive or learning-based alert system
6. **Not Investment Advice**
    - Outputs are informational only
    - Should not be used for financial decision-making

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10. Future Improvements

Planned enhancements include:

1. **Automated Data Ingestion**
    - Bulk upload support
    - Automatic report type detection
2. **Advanced Risk Classification**
    - Fine-tuned NLP models
    - Categorization of regulatory, financial, and operational risks
3. **Vector Search Over Reports**
    - Semantic search across historical filings
    - Faster retrieval of relevant sections
4. **Multi-Report Comparison**
    - Year-over-year filing comparison
    - Sentiment and risk trend tracking

5. **Enhanced Financial Analysis**
    - Financial ratio computation
    - Forecasting and scenario analysis
6. **Intelligent Alerting**
    - ML-driven alert generation
    - Pattern learning from historical data
7. **Scalability & Deployment**
    - Containerization for cloud deployment
    - Performance optimization for large datasets
8. **Advanced Visualization**
    - Interactive drill-downs
    - Exportable reports (PDF/CSV)